

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 20 August 2026, 8:47 AM**State** Finished**Completed on** Thursday, 20 August 2026, 8:48 AM**Time taken** 1 min 36 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int main() {
3      int n, count = 0;
4      if (scanf("%d", &n) != 1) return 0;
5      int rev = 0, remainder;
6      count++;
7      while (n != 0) {
8          count++;
9          remainder = n % 10;
10         count++;
11         rev = rev * 10 + remainder;
12         count++;
13         n /= 10;
14         count++;
15     }
16     count++;
17     count++;
18     printf("%d", count);
19     return 0;
20 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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